

PQC Migration Readiness Assessment Report

Governance, Business Impact, and Regulatory Migration Intelligence

Assessment Target	Streple Backend
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Report Type	Executive Security Assessment
Prepared By	PQC Migration Engine

Executive Summary

This report summarizes post-quantum cryptographic migration readiness, business impact, regulatory exposure, governance status, and recommended migration urgency based on the scanned application codebase.

Metric	Value
Affected Systems / Findings	12
High Priority Findings	3
Medium Priority Findings	6
Low Priority Findings	3
Average Readiness Score	59.58/100
DORA High Impact Findings	2
PCI DSS High Impact Findings	8
Elevated Compliance Risk Findings	3
Recommended Migration Deadline	2028

Business Impact Summary

Asset	Business Zone	Throughput	Latency	Criticality
JWT Authentication Strategy	Customer Authentication	Medium Throughput	Elevated	High
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Public Key Handling	API Gateway	Medium Throughput	Elevated	High
Digital Signature Verification	API Gateway	Medium Throughput	Elevated	High
JWT Authentication Service	Customer Authentication	Medium Throughput	Elevated	High
JWT Authentication Service	Customer Authentication	Medium Throughput	Elevated	High
RSA-OAEP Encryption	Critical Payments	High Throughput	Critical	Critical
Public Key Encryption	Critical Payments	High Throughput	Critical	Critical
Cryptographically Secure Random Generation	Internal Operations	Low Throughput	Normal	Medium
JWT Authentication Strategy	Customer Authentication	Medium Throughput	Elevated	High
Cryptographically Secure Random Generation	Internal Operations	Low Throughput	Normal	Medium
Cryptographically Secure Random Generation	Internal Operations	Low Throughput	Normal	Medium

Regulatory Impact Assessment

Asset	DORA	PCI DSS	NIST PQC	Risk	Deadline
JWT Authentication Strategy	Medium	High	Partial	Moderate	2032
JWT Authentication Strategy	Medium	High	Partial	Moderate	2032
Public Key Handling	Medium	High	Partial	Moderate	2032
Digital Signature Verification	Medium	Medium	Partial	Elevated	2030
JWT Authentication Service	Medium	High	Partial	Moderate	2032
JWT Authentication Service	Medium	High	Partial	Moderate	2032
RSA-OAEP Encryption	High	High	Partial	Elevated	2028
Public Key Encryption	High	High	Partial	Elevated	2028
Cryptographically Secure Random Generation	Low	Medium	Strong	Low	2035
JWT Authentication Strategy	Medium	High	Partial	Moderate	2032
Cryptographically Secure Random Generation	Low	Medium	Strong	Low	2035
Cryptographically Secure Random Generation	Low	Medium	Strong	Low	2035

Top Priority Migration Actions

Asset	Owner	Phase	Recommendation
Digital Signature Verification	Security Team	Phase 1	Evaluate migration path toward ML-DSA or SLH-DSA.
RSA-OAEP Encryption	Security Team	Phase 1	Evaluate hybrid RSA + ML-KEM deployment.
Public Key Encryption	Security Team	Phase 1	Review public key encryption dependencies and PQC alternatives.

Detailed Technical Findings

Asset	Priority	Coverage	Owner	Status	Recommendation
JWT Authentication Strategy	Medium	Not Yet Assessed	Application Team	Under Review	Assess authentication infrastructure for PQC readiness.
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Public Key Handling	Medium	Not Yet Assessed	Infrastructure Team	Under Review	Review certificate and key lifecycle management for crypto agility.
Digital Signature Verification	High	Horizon Short	Security Team	Planned	Evaluate migration path toward ML-DSA or SLH-DSA.
JWT Authentication Service	Medium	Not Yet Assessed	Application Team	Under Review	Review signing algorithms and key management dependencies.
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RSA-OAEP Encryption	High	Horizon Short	Security Team	Planned	Evaluate hybrid RSA + ML-KEM deployment.
Public Key Encryption	High	Horizon Short	Security Team	Planned	Review public key encryption dependencies and PQC alternatives.
Cryptographically Secure Random Generation	Low	Covers	Platform Team	Monitor	Monitor but typically not a direct PQC migration target.
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